

# Relationship between serum parathyroid hormone levels, vitamin D sufficiency, and calcium intake.

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## Evidence abstract

**CONTEXT** Adequate vitamin D status for optimum bone health has received increased recognition in recent years; however, the ideal intake is not known.

Serum 25-hydroxyvitamin D is the generally accepted indicator of vitamin D status, but no universal reference level has been reached.

**OBJECTIVE** To investigate the relative importance of high calcium intake and serum 25-hydroxyvitamin D for calcium homeostasis, as determined by serum intact parathyroid hormone (PTH).

**DESIGN, SETTING, AND PARTICIPANTS** Cross-sectional study of 2310 healthy Icelandic adults who were divided equally into 3 age groups (30-45 years, 50-65 years, or 70-85 years) and recruited from February 2001 to January 2003.

They were administered a semi-quantitative food frequency questionnaire, which assessed vitamin D and calcium intake.

Participants were further divided into groups according to calcium intake (<800 mg/d, 800-1200 mg/d, and >1200 mg/d) and serum 25-hydroxyvitamin D level (<10 ng/mL, 10-18 ng/mL, and >18 ng/mL).

**MAIN OUTCOME MEASURE** Serum intact PTH as determined by calcium intake and vitamin D.

**RESULTS** A total of 944 healthy participants completed all parts of the study.

After adjusting for relevant factors, serum PTH was lowest in the group with a serum 25-hydroxyvitamin D level of more than 18 ng/mL but highest in the group with a serum 25-hydroxyvitamin D level of less than 10 ng/mL.

At the low serum 25-hydroxyvitamin D level (<10 ng/mL), calcium intake of less than 800 mg/d vs more than 1200 mg/d was significantly associated with higher serum PTH ( $P = .04$ ); and at a calcium intake of more than 1200 mg/d, there was a significant difference between the lowest and highest vitamin D groups ( $P = .04$ ).

**CONCLUSIONS** As long as vitamin D status is ensured, calcium intake levels of more than 800 mg/d may be unnecessary for maintaining calcium metabolism.

Vitamin D supplements are necessary for adequate vitamin D status in northern climates.

## Document metadata

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